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Drug Abuse and the Internet: Evidence from Craigslist

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Abstract. The United States is in the midst of a drug overdose epidemic. Although the online availability of drugs has been a growing concern with considerable speculation that digital platforms are contributing to this epidemic, empirical assessments have been lacking. To quantify this impact, we rely on the phased rollout of Craigslist, a major online platform, as an experimental setup. Applying a difference-in-differences approach on a national panel data set for all counties in the United States from 1997 to 2008, we find a 14.9% increase in drug abuse treatment admissions, a 5.7% increase in drug abuse violations, and a 6.0% increase in drug overdose deaths after Craigslist's entry. The impacts of Craigslist's entry are larger among women, whites, Asians, and the more educated. Further, the unintended consequences of Craigslist are more likely to accrue in larger, wealthier areas with initially low levels of drug abuse. These findings raise the possibility that the marked growth in U.S. drug abuse may have partially stemmed from the wider availability of illicit drugs online at the very beginning of its evolution.

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Keywords: public health • drug abuse • online platforms • natural experiment

1. Introduction

In 2017, more than 70,000 people died from drug overdoses, surpassing the number of deaths from HIV, gun violence, or traffic accidents at their peaks (Ahmad et al. 2018). In response to this growing crisis, the Food and Drug Administration (FDA) convened a special summit in June 2018,¹ bringing to the table central internet stakeholders with the goal of “combating the illicit online sale of controlled substance.”

However, the warning that the internet has been a major culprit for drug diversion and abuse is not new. The earliest report in the medical literature describing the sale of illicit drugs through the internet can be traced back to 2001 (Lieberman 2001). Since then, media articles covering the wide availability of drugs online has surged. For example, in 2003, a *Washington Post* article on drug trafficking suggested that “the internet has become a pipeline for narcotics and other deadly drugs” (Gaul and Flaherty 2003). In 2005, the Drug Enforcement Agency administrator asserted that “the internet has become an open medicine cabinet, a help-yourself pill bazaar to make you feel good” (Kaufman 2005). In 2007, a report submitted to the Senate Judiciary Committee stated that “the internet has become a pharmaceutical candy store stocked with addictive drugs, available at the click of a mouse to any kid with a credit card number” (Califano 2007). Despite the evidence that online platforms have

increased the accessibility of drugs to wider sections of the population, what is less clear is the degree to which these platforms are directly responsible for the increase in drug abuse and whether they have had disproportionate effects on different segments of the population, categorized by gender, race, income, and other demographic factors.

Academic evidence to date has also been largely anecdotal, and the findings are sometimes contradictory. In a systematic internet-based monitoring study, Forman et al. (2006) documented an abundance of websites offered opioids without prescriptions, whereas studies relying on survey data have consistently shown that the internet to be only a *minor* source for illicit purchase of drugs (Orizio et al. 2011). Within this context, we aim to provide the first empirical evidence of the impact that a major internet platform that connects buyers and sellers for a variety of products and services has on the incidence of drug abuse when it enters new markets. To this end, we rely on the phased rollout of Craigslist as an experimental setup.

Craigslist is one of the largest online platforms with classifieds devoted to jobs, housing, personals, and goods and services for sale. Despite its many positive features, Craigslist has also served to facilitate numerous illicit activities, including prostitution (Chan et al. 2018), risky sexual behavior (Greenwood and Agarwal 2015), and in particular, illicit drug use

(Tofighi et al. 2016). The platform was first established in San Francisco in 1995 and started expanding to other U.S. cities in 2000. During its expansion, it was made available in different locations in a staggered fashion across the years. Given that each Craigslist site is location-specific, the drug ads posted on the platform cater to the population living within the location. This unique entry setting allows for the econometric comparison of “treated” areas with “control” areas.

Applying a difference-in-differences approach on a national panel data set, our analysis reveals that the entry of Craigslist in a geographic area is associated, on average, with a 14.9% increase in drug abuse treatment admissions, 5.7% increase in drug abuse violations, and 6.0% increase in drug overdose deaths. In addition, the impacts of Craigslist are not limited to existing abusers; in fact, there is a more dramatic increase in the number of new abusers. Analysis on subsamples shows that the impacts of Craigslist entry are larger among women, whites, Asians, and the more educated—populations that are traditionally considered to be at lower risk of drug abuse. Further, the unintended consequences of Craigslist are more likely to be observed in wealthier areas with higher incomes and lower unemployment rates. These findings lend empirical credence to the argument that the drug overdose crisis in the United States may, in part, be attributable to the wider availability via online platforms.

2. Theoretical Mechanisms

I bought drugs online for years I liked the idea of anonymously making quick drug deals with people in my area. I didn't have to call around to a bunch of shady people to find out who had what I was looking for, and I didn't have to argue about the price. I could simply reply to the ad that fit my needs.²

There are many characteristics of a substance that influence the degree to which it is abused, ranging from its onset of action to the intensity of the euphoric experience. One of the most influential characteristics however is its *availability* (Anderson and Hughes 2000, Lieberman 2001). For example, the most available substances of abuse—tobacco and alcohol—cause the greatest public health problems. Illicit opioids, such as heroin, however, have generally been available only through illegal channels, which has restricted their use to those who are willing to undergo the risk of arrest or potential exposure to the violence associated with the illegal drug trade (Lieberman 2001). Support for this argument can also be found in the case of marijuana legalization, with which higher availability increases the likelihood of abuse (Wang et al. 2016, Hasin et al. 2017), and the increasing availability of low-cost synthetic opioids (e.g., fentanyl),

which might have further fueled the ongoing opioid epidemic (Darcq and Kieffer 2018). Historically, illicit drug retail markets have operated in physical spaces with associated practical limitations and boundaries. The internet has emerged as a popular platform that lowers transaction costs by facilitating buyer–seller matching (Bailey and Bakos 1997) and reducing information asymmetry in markets (Williamson 1981, Brynjolfsson et al. 2003). Simply by making it easier to find and buy drugs online, internet platforms encourage more consumption of drugs.

An additional advantage provided by the internet is *anonymity*. It is well documented that people may behave differently under the cloak of anonymity. Comparing the tone of speech across thousands of anonymous and nonanonymous posts, Santana (2014) establishes the link between rampant incivility and anonymity. Through repeated social dilemma experiments, Wang et al. (2017) demonstrate that cooperation is less likely to be achieved under anonymity. Psychology literature suggests that the security of anonymity (i.e., a lower likelihood of being discovered) is associated with more risky behaviors (Rogers 2011). Because drug use has long been perceived as inherently deviant or counter to widespread social norms (Leshner 1997), anonymity enables drug users and peddlers to remain immune to such social stigma. That is, anonymity—though not critical for the market to function—can reduce the effect of social frictions that might otherwise inhibit illicit drug transactions.

By making illegal drugs more accessible and convenient to buy and less risky because of anonymity, the internet encourages greater consumption of drugs. From a supply-side perspective, online drug sales are likely to be even more lucrative as the costs and risks of online transactions are lower than off-line sales. Therefore, the entry of online platforms is likely to increase market participation by improving liquidity and spurring both demand and supply of illicit drugs.

Although overall illegal trade in drugs is expected to increase as a result of online platform entry, it is not obvious as to which subpopulations would be more affected. On the one hand, risky behaviors are known to be more prevalent among people of lower socioeconomic status (SES) (Sampson et al. 1997). In particular, illicit drug users can be characterized by poor family relationships, high peer substance abuse, living in poverty, low socioeconomic and educational status, and high index of novelty- or sensation-seeking behavior (Jekel and Allen 1987). It is, therefore, natural to hypothesize that people of lower SES, who are more likely to have prior experience with drugs, may be spurred to consume more because of its wider availability. Another line of reasoning, however, may suggest the opposite prediction. Although risky

behaviors generally have a lower rate of acceptability among people of higher SES (Paul et al. 2010), this trend could be changed by the cloak of anonymity. As noted by Greenwood and Agarwal (2015), the entry of online matching platforms in communities increases HIV incidence through the solicitation of casual sex and, most strikingly, the greatest absolute increase occurs among populations of higher SES. Also, the socioeconomically advantaged have higher digital access and capabilities with regard to use of online resources; that is, they are far more skillful in their exploitation of online resources (Norris 2001, Po-An Hsieh et al. 2011). These lead to the possibility that people of higher SES are more likely to be enticed by the illicit drug sales facilitated by the internet. Our empirical analyses aim to tease out these contrasting predictions and shed light on population characteristics that exhibit greater vulnerability to the sale of online drugs.

3. Data and Methodology

To empirically examine the relationship between online drug availability and its abuse, we constructed a unique data set from various sources. An indicator variable was created for the presence of Craigslist,³ which serves as the main independent variable of interest.

To measure drug abuse, we constructed three variables from different sources. The primary measure was drug abuse treatment admissions, obtained from Treatment Episode Data Set-Admissions (TEDS-A), maintained by the Substance Abuse & Mental Health Data Archive. TEDS is a national data system designed to track admissions into all substance treatment facilities that receive federal funding, including approximately 83% of all eligible drug or alcohol treatment admissions in the United States. We obtained TEDS data from 1997 to 2008. For each year, we observed patient-level data, including demographics (e.g., age, gender, race, education level, etc.), geographical information (e.g., census state Federal Information Processing Standards (FIPS) code, FIPS 2000 core-based statistical area (CBSA) code),⁴ substances used by the patient, number of prior treatment episodes, and service type (e.g., detoxification, long-term rehabilitation, etc.). Drug abuse treatment admissions were defined as patients who identified drugs as the only substance of abuse reported at the time of admission to treatment. We aggregated the drug abuse treatment admissions at the CBSA level for each year from 1997 to 2008. To investigate the effect of Craigslist introduction across different subpopulations, we then aggregated our dependent variable based on sociodemographic indicators, such as age, gender, race, and education. In particular, we considered two subgroups:

individuals admitted into the detoxification program with no prior treatments (defined as “new abusers”) and those with at least one prior treatment (defined as “existing abusers”).

In addition, we used annual county-level drug abuse violations from the FBI’s Uniform Crime Report (UCR) database as our secondary measure.⁵ The tertiary measure of drug abuse was drug overdose deaths,⁶ collected from the multiple cause of death data field in the Wide-Ranging Online Data for Epidemiologic Research (WONDER) database maintained by the Centers for Disease Control and Prevention (CDC),⁷ in which the first year available is 1999.

To account for potential confounding factors that might influence drug abuse and Craigslist entry in each CBSA, we included annual county-level demographic characteristics, socioeconomic factors, crime-related trends, and health services from various sources as control variables. Specifically, the population size, demographic decomposition (e.g., age group proportions, ethnicity proportions), and poverty data were obtained from the U.S. Census Bureau. Income and unemployment data were obtained from the Bureau of Labor Statistics, and crimes and arrests data from the Federal Bureau of Investigation. Each of these variables was provided at the county level for each year. In addition, we identified county-level changes in health services using data from the U.S. Census Bureau’s County Business Patterns.⁸ Finally, we merged the aforementioned data and obtained a national panel data set on the CBSA level for drug abuse treatment admissions (1997–2008) and on the county level for drug abuse violations (1997–2008) and drug overdose deaths (1999–2008). Table 1 provides the descriptive statistics of our data for the main analysis.

Our statistical method used a difference-in-differences model, specified as follows:

$$\ln(\text{DrugAbuse}_{it}) = \delta_i + \delta_t + \omega_i t + \alpha \cdot X_{it} + \beta \cdot \mathbb{I}(\text{entry}_{it}) + \xi_{it}, \quad (1)$$

where DrugAbuse_{it} is the dependent variable reflecting drug abuse level in CBSA i , year t , with log transformation used to reduce its skewness; $\mathbb{I}(\text{entry}_{it})$ is the binary indicator for Craigslist’s entry,⁹ which equals one for all of the years following Craigslist’s entry and zero otherwise; δ_i is a vector of CBSA fixed effects and δ_t is a vector of year fixed effects; $\omega_i t$ is a CBSA-specific linear time trend, controlling for *unobservable* CBSA-level factors that evolve over time; X_{it} represents a vector of control variables, accounting for *observable* time-varying CBSA-level factors, such as population, income, and unemployment level; and ξ_{it} is the error term. Note that, in this specification, the fixed effects help to compare drug abuse incidences

Table 1. Descriptive Statistics

Variables	Observations	Mean	Standard deviation	Minimum	Maximum
log(<i>Drug Abuse Treatment Admissions</i>)	3,453	7.578	1.969	0.693	15.124
log(<i>Population</i>)	3,453	12.809	1.032	11.437	16.778
<i>Male Proportion</i>	3,453	0.493	0.011	0.467	0.573
<i>White Proportion</i>	3,453	0.857	0.113	0.219	0.991
<i>Age 15–29 Proportion</i>	3,453	0.206	0.032	0.106	0.388
<i>Age 65+ Proportion</i>	3,453	0.132	0.033	0.052	0.354
<i>Income (k)</i>	3,441	42.460	8.690	20.034	84.250
<i>Unemployment Rate</i>	3,453	5.227	1.852	1.200	22.600
<i>Poverty Rate</i>	3,427	12.910	5.084	2.500	40.700
<i># of Substance Abuse Treatment Facilities (per 100,000 people)</i>	3,453	4.808	3.178	0.000	34.385
<i>Total Crime Rate</i>	3,403	0.038	0.014	0.008	0.130
<i>Total Arrest Rate</i>	3,453	0.045	0.021	0.000	0.154

across CBSAs and time periods, and the coefficient β is the difference-in-differences estimate of the effect of Craigslist's entry on the total admissions.

4. Results

Overall, we observe a strong and significant increase in the number of patients admitted for drug abuse treatment (see Table 2). After Craigslist's entry, the total admissions of drug abuse treatment increase by 14.9% per CBSA year, the magnitudes and significance of the effect are consistent across all models.

Further analysis reveals that different subpopulations are disproportionately affected by the platforms (see Figure 1).¹⁰ For example, women are more susceptible to the influence than men, and after Craigslist entry, the positive effect on drug abuse is the greatest among the middle-aged (between 45 to 54 years). The impacts of Craigslist are also heterogeneous across races: compared with the effect on black people, the impacts are larger among the white population and even larger among Asians. Although every education level is impacted by the entry of

Craigslist, the effect is more pronounced for people of higher education, especially those with a high school or an associate degree. In addition, compared with the impact on existing abusers, there is a greater and significant increase in new abusers. Collectively, these observations suggest that, by and large, the unintended consequences of Craigslist are more likely to be observed among the socioeconomically advantaged (e.g., the white population, the more educated) or populations that might be more afraid of social stigma attached to consuming drugs (e.g., women, Asians, and the middle-aged). These findings also partly explain the significant increase in new drug abusers.

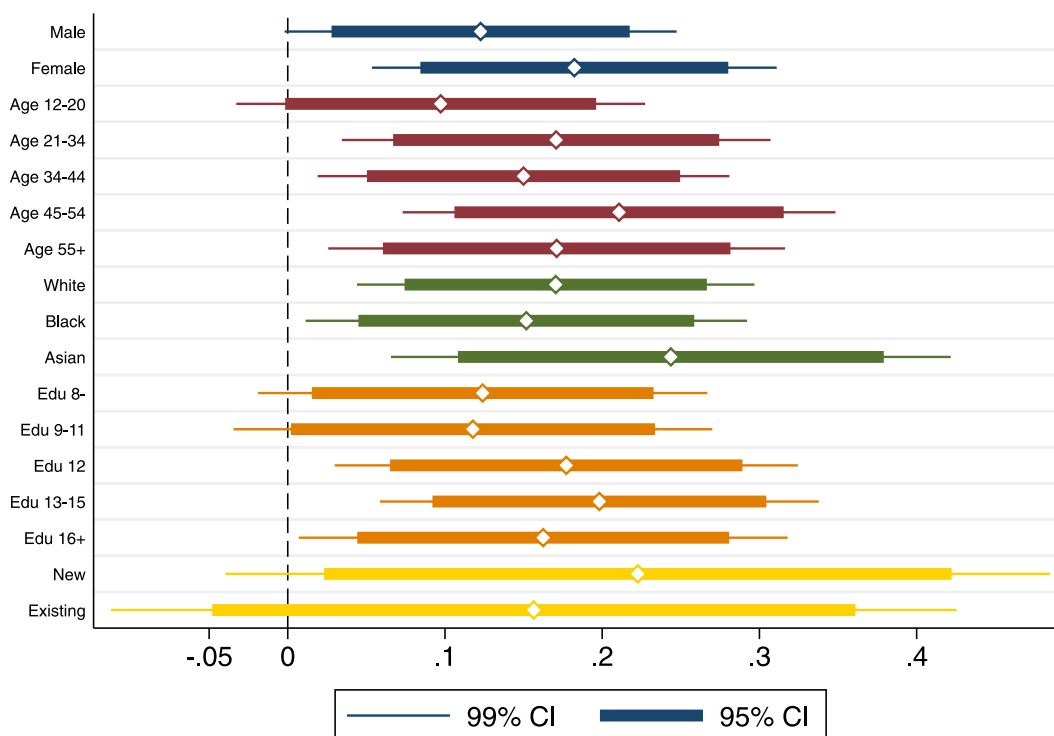
We then partitioned the data into four subsets based on the quantile points for each socioeconomic factor (e.g., population, income, crime rate, etc.) and repeated the analysis on each of the subsets. Figure 2 shows that the increasing trends of drug abuse after Craigslist's entry are more likely to be observed in areas of larger population (third and fourth quantiles), higher proportion of white people (last three quantiles), higher incomes (last three quantiles), and lower

Table 2. Impact of Craigslist Entry on Drug Abuse Treatment Admissions

	(1)	(2)	(3)	(4)	(5)
Craigslist	0.152*** (0.047)	0.149*** (0.048)	0.149*** (0.047)	0.149*** (0.046)	0.149** (0.067)
CBSA fixed effects	✓	✓	✓	✓	✓
Year fixed effects	✓	✓	✓	✓	✓
CBSA linear time trend	✓	✓	✓	✓	✓
Controls	—	✓	✓	✓	✓
Robust standard errors	—	—	✓	—	—
Bootstrapped standard errors	—	—	—	✓	—
Clustered standard errors	—	—	—	—	✓
N	3,452	3,371	3,371	3,371	3,371
Adjusted R ²	0.928	0.929	0.929	0.929	0.929
Log pseudolikelihood	−2,290.3	−2,223.6	−2,223.6	−2,223.6	−2,223.6

Note. Standard errors are shown in parentheses.

* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

Figure 1. (Color online) Impact of Craigslist on Drug Abuse Treatment Admissions for Different Subgroups

Notes. The x -axis shows the estimated effects of Craigslist using Equation (1) on difference subgroups along the y -axis. The 99% and 95% confidence intervals were depicted by thick and thin bars, respectively.

unemployment (first and second quantiles). Furthermore, the unintended consequences of Craigslist tend to accrue in areas with initially the lowest level of drug abuse violations (first and third quantiles), and treatment admissions (first and second quantiles). Consistently, areas with a lower level of drug abuse treatment facilities (first and third quantiles) are more susceptible to the impacts. Clearly, these areas are in more severe need of expanding the treatment-related resources after Craigslist's entry. Finally, the impacts of Craigslist are more notable in areas with a higher electronic shopping level (third and fourth quantiles), which suggests that areas with higher digital access are more likely to be victims of the platform entry.

Despite the established links between low-income neighborhood and risky behaviors (Sampson et al. 1997, Kling et al. 2007), our results suggest that the more affluent areas, in which crimes and drug abuse were less prevalent prior to the platform entry, are more susceptible to the impact of Craigslist. In contrast, the residents of relatively disadvantaged neighborhoods are less likely to be affected by the platform entry.

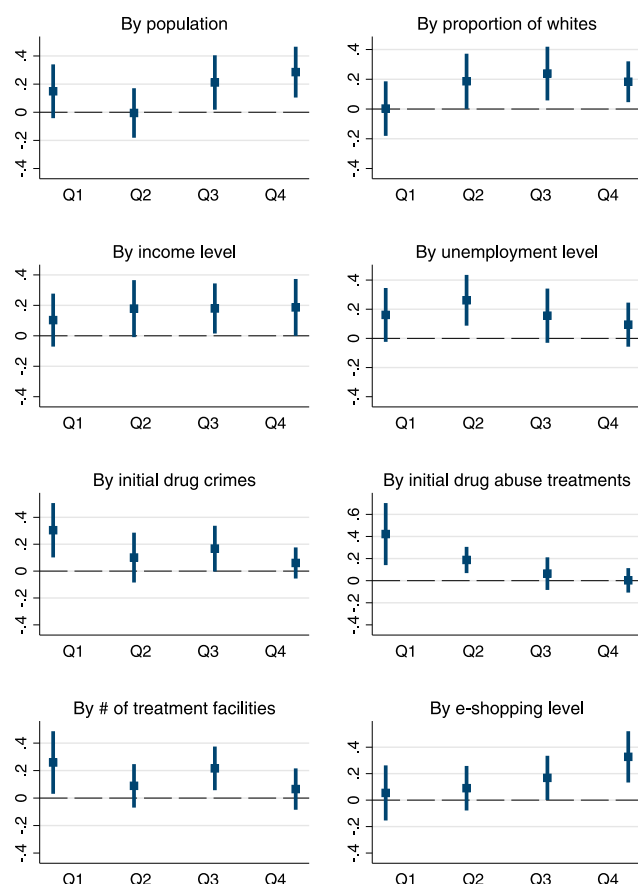
Complementary to the main findings, we analyzed annual county-level arrests for drug abuse violations between 1997 and 2008 as our secondary evidence. The results show that, after Craigslist's entry, drug abuse violations increase by 5.7% per county year. In addition, analysis on CDC Multiple Cause of Death

data suggests that, on average, drug overdose deaths increase by 6.0% in counties where Craigslist entered.

5. Robustness Checks

We conducted a series of robustness checks to further ensure the validity of the main results. First, an assumption underlying our difference-in-differences strategy is that treated and control CBSAs were similar prior to the entry of Craigslist. We assessed this assumption by estimating the effect of Craigslist entry within an event study framework; it can rule out the existence of linear trends from leading years and subsequent years and, thus, provide a sense of long-run impact of the intervention. Second, to derive a more comparable set of CBSAs for analysis, we used propensity score matching (PSM) to select CBSAs with and without Craigslist entry that are statistically similar in terms of demographic, socioeconomic, and crime indicators. Third, one of the main assumptions for our estimated effects to be causal is that the entry of Craigslist was exogenous with respect to drug abuse patterns. The validity of this assumption was assessed by a Cox proportional hazard model, which predicted the platform entry as a function of drug abuse and other independent variables. Finally, we conducted two falsification checks to ensure that our results are not idiosyncratic. Our main findings are robust to all of the preceding tests.

Figure 2. (Color online) Impact of Craigslist on Drug Abuse Treatment Admissions by Geographic Area



Notes. Data were partitioned into Q1, Q2, Q3, and Q4 based on the 25th, 50th, and 75th quartile points for each socioeconomic factor. For each subfigure, the estimated β from Equation (1) and its 95% confidence interval are shown.

5.1. Event Study Framework

The relationship between Craigslist's entry and drug abuse may be driven by unobserved confounding factors that existed prior to Craigslist's entry. To assess this possibility, we evaluate the effect of Craigslist's entry within an event study framework. This method reorients each CBSA's first posttreatment period to time zero and measures the difference between treated and control CBSAs in the periods prior to and following this period. Specifically, we estimate the following equation:

$$\ln(\text{DrugAbuse}_{it}) = \delta_i + \delta_t + \alpha \cdot X_{it} + \sum_h \beta_h T_{ih} + \xi_{it}, \quad (2)$$

where $T_{ih} = 1$ if CBSA i had Craigslist h years ago (or will have Craigslist $-h$ years in the future when $h < 0$); β_h represents the difference between the treatment and control CBSAs in period h as compared with the last pretreatment period ($h = -1$).

This analysis serves two purposes. First, systematic differences between treated and control counties

in the pre-entry period could suggest nonparallel trends or endogenous entry decisions, which would violate the assumption of the difference-in-differences approach. The relative time model as specified is a standard way to test the potential for nonparallel trend (Angrist and Pischke 2008) and has been widely used in extant literature dealing with problems within similar contexts (Chan and Ghose 2014, Greenwood and Agarwal 2015, Burtch et al. 2018). Second, it can provide a sense of long-run impact of the intervention, namely whether the impact of Craigslist entry fades out or grows over time.

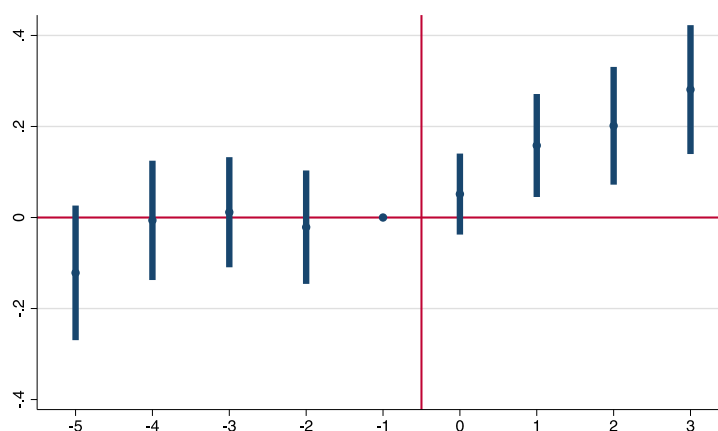
Figure 3 presents the full set of coefficients from estimating our baseline event-study specification (Equation (2)) for drug abuses. First note that the time dummies are not significantly different from zero before the entry of Craigslist. This suggests that there are no systematic differences between treated and control states in the pretreatment period, and therefore, the assumption of the difference-in-differences model was not being violated. Following the platform's entry, there is a sudden increase in drug abuse outcome, and the magnitude of the effect grows larger and larger in the posttreatment years, suggesting that the effect is likely to grow in the long run.

5.2. Difference-in-Differences with Propensity Score Matching

The parallel trends assumption underlying the difference-in-differences technique can also be violated in the absence of an appropriate control group. A matching approach can reduce the bias between the treated and control groups, thereby allowing for more accurate comparisons of the drug abuse trends *before* and *after* Craigslist's entry (Heckman et al. 1997). Specifically, we used PSM to generate a set of control CBSAs that are statistically similar to the treated CBSAs with respect to demographic, socioeconomic, and crime indicators.

We first used the nearest neighbor matching algorithm to identify the closest 3, 5, and 10 control units, respectively, for each treated unit. Matching with a caliper of 0.001, we found that more than 90% of the treated units were able to be successfully matched. To further ensure that our sample balance had improved, we compared the differences in means (before versus after) between treated and control units for the observed characteristics. After matching, the differences became insignificant across all variables. Figure 4 shows that, after matching, the propensity score distribution became much closer between the treated and control counties. This suggests that the parallel trends assumption, which underlies the difference-in-differences approach, is satisfied.

We then applied the difference-in-differences approach on the matched sample, and the impacts of

Figure 3. (Color online) Impact of Craigslist on Drug Abuse Treatment Admissions Over Time

Notes. For each period, the estimated β_h from Equation (2) and its 95% confidence interval are shown. The estimation signifies the difference between treated and control CBSAs h periods before and after Craigslist's entry, net of CBSA and year fixed effects, and relative to last pretreatment period ($h = -1$).

Craigslist were positive and significant. Using alternative matching algorithms (i.e., radius matching and kernel matching), the results were consistent (see Table 3).

5.3. Entry Exogeneity

One of the main assumptions for our estimated effects to be causal is that the entry of Craigslist was exogenous with respect to drug abuse patterns. Although there is no reason to suspect that Craigslist's entry into a given geographic region was influenced by the level of drug abuse in that region, we cannot rule out the possibility that some unobserved factors related to drug abuse might have induced Craigslist's entry. To assess this possibility, we used a Cox proportional hazard model, which predicted the platform entry as a function of drug abuse and other independent variables (Cox 1972). The coefficients of drug abuse would provide insights into the possibility of reverse causality. Note that our survival time was measured in years, which may result in an issue of *tied events*, that is, events with exactly the same survival time. This issue was resolved by three approximations for handling ties, that is, the Breslow, Efron, and exact marginal-likelihood methods, respectively (Hosmer et al. 2008).

The results consistently show that the platform's decision to enter a geographic region is influenced by certain demographic and socioeconomic factors (see Table 4). For example, larger populations and higher incomes are correlated with an increased likelihood of Craigslist's entry. These factors typically determine its potential usage after launch as noted by Chan and Ghose (2014), Dhanorkar (2018), and Greenwood and Agarwal (2015). However, there is no evidence of reverse causality: the coefficients of *drug_abuse* are not

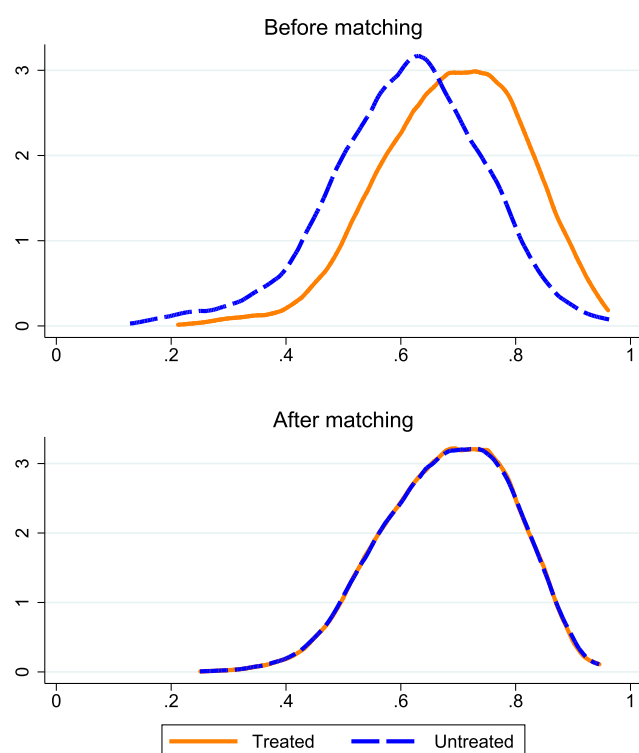
significant across the three models, which indicates that the drug abuse level in a given area is not predictive of Craigslist's entry.

5.4. Falsification Checks

Finally, we conducted two falsification checks to ensure that our results were not idiosyncratic. In the first test, we randomized the entry of Craigslist and repeated the analysis of our main model. This test was carried out to determine how likely it was that a random implementation of Craigslist would yield an aggregate effect size comparable to our estimates (Burtch et al. 2013, Greenwood and Agarwal 2015, Greenwood and Wattal 2017). To execute this analysis, we randomly assigned Craigslist's pseudo-presence to all counties and treated counties, respectively. We then estimated the effect of the random entry of Craigslist using the difference-in-differences model as specified in Equation (1) to obtain the coefficient for the pseudo-treated (denoted as β_{random}). This procedure was repeated 1,000 times for each of the two random assignments. Table 5 summarizes the mean and standard deviation of the estimated β_{pseudo} , and the Z-score of the difference between the estimated β_{pseudo} and our original estimate β . In the random assignments, the average estimation of pseudo-treated (β_{pseudo}) were 0.0001 and 0.0005, respectively, and our estimated effect size (β) is significantly larger than what might have been expected purely by chance (β_{pseudo}). This indicates that our estimation is not spurious, and there is no significant unobserved difference between treated and control counties.

In the second test, we repeated our analysis using alcohol abuse treatment admissions, other crimes, and deaths of other underlying reasons as our dependent

Figure 4. (Color online) Propensity Score Before and After Matching



Notes. The propensity score indicates the probability of Craigslist's entry given the values of all potential confounders. It was estimated by a probit regression model with the all confounders as predictors.

variables. As expected, the entry coefficients did not pick up any effect in the falsification regressions. The results are shown in the online appendix.

6. Discussions

The online sale of drugs started in the late 1990s, and this illegal activity has been growing with the expansion of the internet. It first appeared on website forums before moving to search engine results and is now populating social media websites, such as Facebook, Twitter, and Instagram while also appearing on the Dark Web. In this study, we analyzed the impact of

Craigslist on drug abuse violations, treatment admissions, and drug overdose deaths at the national level. Using the phased rollout of Craigslist as an experimental setup, our analysis yields three main results.

First, Craigslist's entry is associated with a 14.9% increase in drug abuse treatment admissions, a 5.7% increase in drug abuse violations, and a 6.0% increase in drug overdose deaths. Analysis using an event study framework further indicates that the momentum of increasing drug abuse grew over time. The persistent positive relationship uncovered in this study serves as an important policy input that affirms the soundness of the FDA's efforts to crack down on the flow of illegal drugs online. Second, although economic disadvantages (e.g., unemployment, poverty, and low education) are thought to be associated with higher risk of substance misuse, we find populations that are generally considered to be "lower risk" are more likely to engage in drug abuse after Craigslist's entry. Consistently, analysis by geographic area suggests that the unintended consequences of Craigslist tend to be observed in wealthier areas where crimes and drug abuse were less prevalent prior to the platform entry. These findings provide insights regarding susceptible locations in which platform entry is likely to lead to higher drug abuse cases. This calls for additional law enforcement and healthcare resources to be allocated proactively in previously unaffected areas in anticipation of changes in drug abuse associated with online access. In addition, the impacts are not limited to existing abusers; indeed, there is a more dramatic increase in the number of new abusers. Collectively, these results strongly suggest that, beyond wider availability, the cloak of anonymity provided by online platforms, such as Craigslist, has enabled new populations that were previously unaffected to enter this market. A key policy implication of this result suggests that these communities might also benefit the most from early intervention programs designed to screen out and provide counseling to new users before drug use

Table 3. Difference-in-Differences Estimation After Propensity Score Matching

	Nearest neighbor matching			Radius matching	Kernel matching
	$k = 3$	$k = 5$	$k = 10$		
Craigslist	0.186*** (0.056)	0.204*** (0.053)	0.154*** (0.046)	0.167*** (0.046)	0.217*** (0.052)
CBSA fixed effects	✓	✓	✓	✓	✓
Year fixed effects	✓	✓	✓	✓	✓
CBSA linear time trend	✓	✓	✓	✓	✓
N	2,518	2,864	3,111	3,141	3,361
Adjusted R^2	0.976	0.976	0.974	0.974	0.971
Log pseudolikelihood	-1,000.1	-1,149.2	-1,327.3	-1,326.2	-1,630.6

Note. Robust standard errors are shown in parentheses.

* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

becomes habitual. Finally, our study highlights the need for more proactive regulation of online platforms. Although online platforms were allowed to grow unchecked in the early internet growth period, there have been increasing calls for “duty to care,” which stipulates that providers of online services have the responsibility to anticipate and mitigate harms that their service might cause to users and to the broader society. Many popular platforms, including Google, Amazon, Facebook, YouTube, Uber, and others, have come under greater scrutiny in recent years. Our work adds empirical punch to the growing chorus for stricter regulations and monitoring of online platforms.

Before we conclude this paper, we want to acknowledge some limitations of this study. First, our estimates of Craigslist’s impacts are possibly conservative because the utilization rates of Craigslist for treated counties were not observed. Second, we were unable to capture the effect on people who did not seek treatment. However, unless there is a significant change in patients’ propensity for seeking treatment before and after Craigslist’s entry, this is unlikely to have biased our results. Third, it is possible that other platforms, such as online pharmacies,

Table 4. Cox Proportional Hazard Model for Craigslist’s Entry

	(1)	(2)	(3)
<i>population</i>	2.33e-07*** (6.57e-08)	2.62e-07*** (6.72e-08)	4.09e-07*** (7.44e-08)
<i>p_male</i>	0.126 (6.618)	−0.975 (7.385)	−0.597 (8.067)
<i>p_white</i>	0.544 (0.627)	0.714 (0.719)	0.606 (0.812)
<i>age_15-29</i>	−2.118 (2.087)	−1.946 (2.319)	−2.553 (2.863)
<i>income</i>	0.035** (0.014)	0.036** (0.015)	0.045*** (0.015)
<i>unemployed_r</i>	−0.048 (0.045)	−0.054 (0.051)	−0.055 (0.063)
<i>poverty_r</i>	0.013 (0.019)	0.012 (0.022)	0.014 (0.025)
<i>e_shopping_r</i>	−0.049 (0.038)	−0.055 (0.044)	−0.061 (0.050)
<i>crimes</i>	−3.411 (4.884)	−3.240 (5.436)	−5.590 (7.271)
<i>arrests</i>	1.743 (2.709)	2.293 (3.159)	2.660 (4.179)
<i>drug_abuse</i>	−2.90e-07 (4.17e-07)	−3.19e-07 (4.27e-07)	−8.12e-07 (7.90e-07)
<i>N</i>	2,810	2,810	2,810
Log pseudolikelihood	−1,106.3	−1,076.6	−498.2

Notes. Models (1)–(3) use the Breslow, Efron, and exact marginal-likelihood methods to handle tied failures, respectively. Robust standard errors are shown in parentheses.

* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

Table 5. Randomized Entry Time of Craigslist (Placebo Test)

	Randomization	
	On <i>all</i> counties	On <i>treated</i> counties
Mean of β_{pseudo}	0.0001	0.0005
Std.Dev of β_{pseudo}	0.0239	0.0269
Estimated β	0.1485	0.1485
Replication times	1,000	1,000
Z-score	−196.3	−174.1
p-value	<0.0001	<0.0001

Notes. The randomization refers to the procedure of randomly assigning Craigslist’s pseudo presence to all CBSAs and treated CBSAs, respectively, with 1,000 times of repetition. β_{pseudo} is the coefficient for Craigslist’s pseudo entry, and β is the coefficient for Craigslist’s entry; both were estimated using difference-in-differences model as specified in Equation (1).

are playing a role simultaneously, and thus, our observed effects might not be completely attributable to Craigslist. However, given the well-established identification strategy we are using and that any other platforms would function via the same mechanism, this concern could be reasonably alleviated.¹¹ In addition, future research might quantify the relation between drug sales on the street versus online to understand to what extent the off-line channel has been affected by the platform entry. Finally, despite the non-trivial externalities resulting from drug abuse, our results do not allow us to draw any conclusions with respect to the overall effect of Craigslist on public welfare.

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Endnotes

¹Online Opioid Summits. <https://www.fda.gov/drugs/news-events-human-drugs/online-opioid-summits>. Accessed January 15, 2019.

²Extracted from *thefix.com*, a website for alcohol and drug addiction recovery news and information. <https://www.thefix.com/man-seeking-dope-look-craigslist-drug-market>, accessed December 5, 2018.

³The entry time of Craigslist was made available on its official web-site (<https://www.craigslist.org/about/expansion>. Accessed May 29, 2018). The final year for the analysis is 2008 because accurate Craigslist entry information is unavailable after 2006. This has also been noted by previous studies that exploit the natural experiment of Craigslist in other contexts (Chan and Ghose 2014, Greenwood and Agarwal 2015, Chan et al. 2018, Dhanorkar 2018).

⁴The CBSA is a collective term for both metropolitan and micro-political (metro and micro) statistical areas, that is, geographic entities defined by the U.S. Office of Management and Budget for use by federal statistical agencies in collecting, tabulating, and publishing federal statistics. Each metro or micro area consists of one or more counties anchored by an urban center of at least 10,000 people plus adjacent counties that are socioeconomically tied to the urban center

by commuting. Note that this is the finest geographic area information that is available in the TEDS data.

⁵ UCR is a nationwide, cooperative statistical effort of nearly 18,000 city, university and college, county, state, tribal, and federal law enforcement agencies voluntarily reporting data on crimes brought to their attention.

⁶ Drug overdose deaths were identified using underlying cause-of-death codes from the 10th revision of ICD (ICD-10): X40-X44, X60-X64, X85, and Y10-Y14.

⁷ Multiple Cause of Death 1999–2016 on CDC WONDER. Atlanta, GA: CDC, National Center for Health Statistics. <http://wonder.cdc.gov>.

⁸ Specifically, substance abuse treatment facilities were identified using the following six-digit NAICS codes: 621420, outpatient mental health and substance abuse centers; 622210, psychiatric and substance abuse hospitals; and 623220, residential mental health and substance abuse facilities.

⁹ For the analysis of TEDS-A data, we first identify the cities where Craigslist had entered and then assigned each city to the appropriate CBSA.

¹⁰ Note that here we compared the *relative* effect sizes across models. In the online appendix, we use Poisson regression to get a sense of the *absolute* effects.

¹¹ A formal test has been done to assess the degree to which omitted unobservable factors can possibly explain away the observed positive relation between Craigslist and drug abuse (see Online Appendix 2.2).

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